

PLANE GEOMETRY

General Polygon (n-sides)

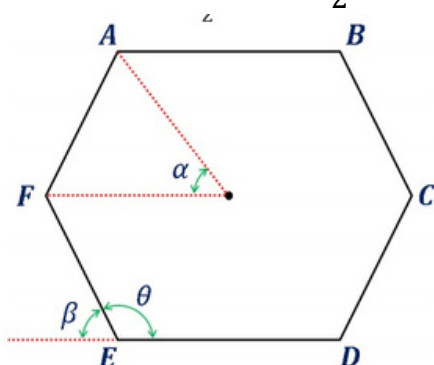
Properties

$$\Sigma \text{Interior Angles } \theta = 180^\circ(n-2)$$

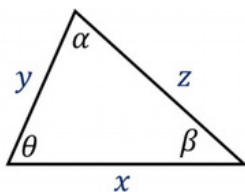
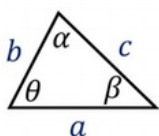
$$\Sigma \text{Exterior Angles } \beta = 360^\circ$$

$$\Sigma \text{Central Angles } \alpha = 360^\circ$$

$$\text{Number of Diagonals} = \frac{n}{2}(n-3)$$



Similar Polygons



$$\frac{a}{b} = \frac{b}{y} = \frac{c}{z}$$

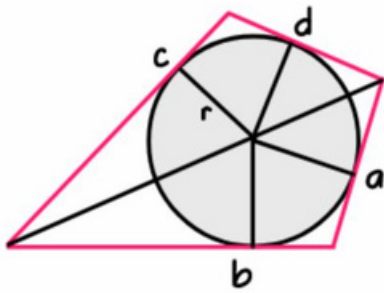
$$\frac{A_{\text{small}}}{A_{\text{big}}} = \left(\frac{a}{x}\right)^2 = \left(\frac{b}{y}\right)^2 = \left(\frac{c}{z}\right)^2$$

Perimeter & Area of Regular Polygons

$$P = ns$$

$$A = \frac{Pa}{2} = \frac{ns^2}{4 \tan\left(\frac{180}{n}\right)}$$

- s = length of one side
- n = number of sides
- a = apothem
o line from the center of a regular polygon



$$A = \sqrt{abcd} = rs$$

Note:

◇ Ptolemy's Theorem

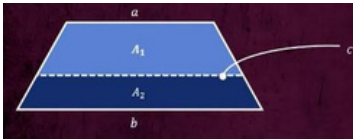
$$ac + bd = d_1 d_2$$

Trapezoids

Area = median x height

$$c = \sqrt{\frac{na^2 + mb^2}{m + n}}$$

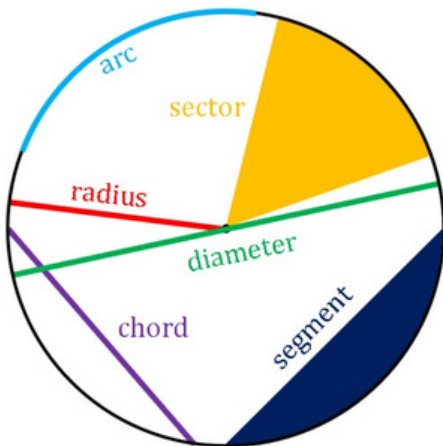
$$\bullet \frac{m}{n} = \frac{A_1}{A_2}$$



Kite & Rhombus

$$A = \frac{1}{2} d_1 d_2 = ah$$

Circles



Sector

$$A_{\text{sector}} = \frac{1}{2} r^2 \theta_{\text{rad}} = \frac{\pi r^2 \theta}{360} = \frac{1}{2} sr$$

Arc Length

$$s = r \theta_{\text{rad}}$$

Segment

$$A_{\text{segment}} = A_{\text{sector}} - A_{\text{triangle}}$$

$$= \frac{1}{2} r^2 (\theta_{\text{rad}} - \sin \theta)$$

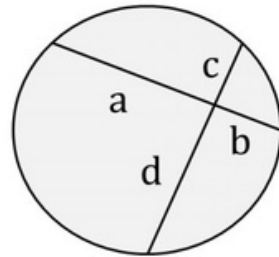


$$\begin{aligned} \text{Area} &= A_{\text{sector}} + A_{\text{triangle}} \\ \text{Area} &= \frac{1}{2} r^2 (\alpha_r + \sin \theta) \end{aligned}$$

where θ , and α , are angles in radians

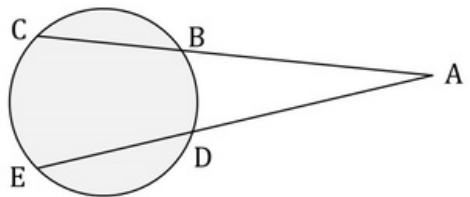
2 Chords Rule

$$\bullet ab = cd$$



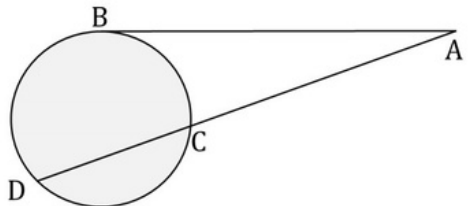
2 Secants Rule

$$\bullet AC \cdot AB = AE \cdot AD$$



Secant-tangent Rule

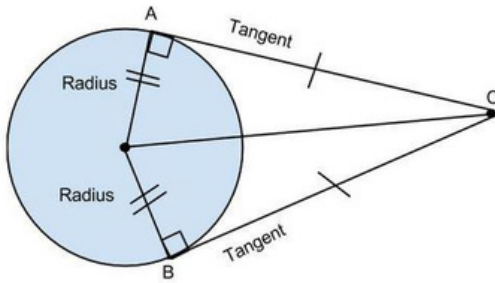
$$\bullet (AB)^2 = AD \cdot AC$$



2 Tangents (PA & PC)

$$\bullet PA = PC$$

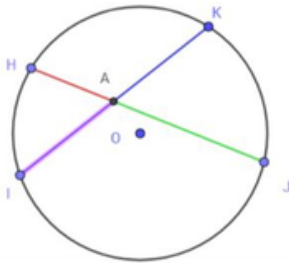
$$\bullet \angle C = \left(\frac{AB_{\text{big}}^2 - AB_{\text{small}}^2}{2} \right)$$



In vs. Out

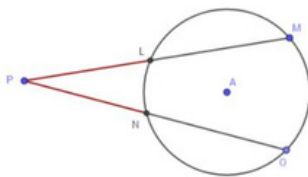
- In

- $\angle A = \frac{HK + I\hat{J}}{2}$
- $\angle HKI = \angle HJI$
- $\angle JHK = \angle JIK$



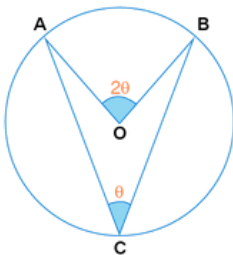
- Out

- $\angle P = \frac{MO + LN}{2}$



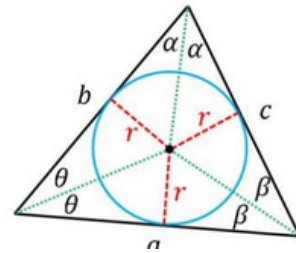
Other Theorems

- Central Angle = 2(Inscribed Angle)



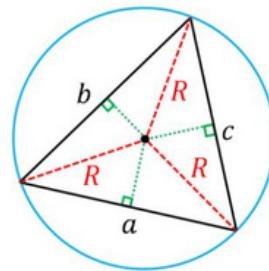
- radius of circle inscribed in a triangle (incenter)

$$r = \frac{A_T}{s}$$



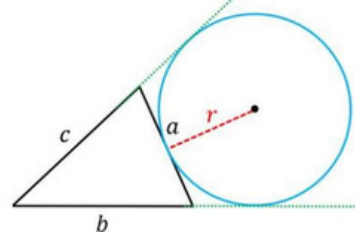
- radius of circle circumscribing a triangle

$$r = \frac{abc}{4A_T}$$

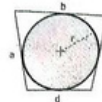


- radius of circle escribed in a triangle

$$r = \frac{AT}{s - a}$$



Circle Inscribed in a Quadrilateral



A circle is inscribed in a quadrilateral if it is tangent to the three sides of the quadrilateral.

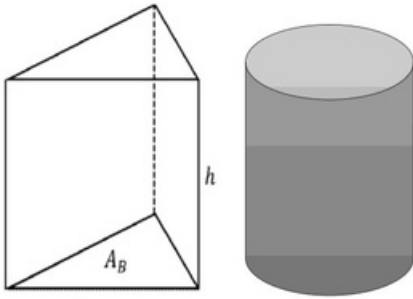
$$r = \frac{A_{quad}}{s}$$

$$A_{quad} = \sqrt{abcd} \quad s = \frac{a+b+c+d}{2}$$

AREA BY APPROXIMATION

The area of any irregular plane figure (such as the one shown) can be found approximately by dividing it into a number of strips or panels by a series of equidistant parallel chords (offsets) $h_1, h_2, \dots, h_n$, the common distance between the chords being d .



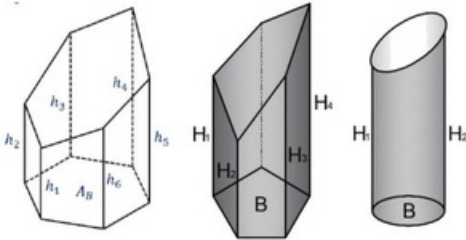


- cylinder

$$V = \pi r^2 h$$

$$TSA = 2\pi r^2 + 2\pi r h$$

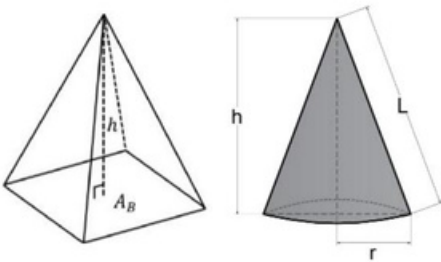
Truncated Prism



$$V = A_B h_{ave}$$

$$TSA = A_1 + A_2 + LSA$$

Pyramids and Cones



$$V = \frac{1}{3} A_B h$$

$$TSA = A_B + LSA$$

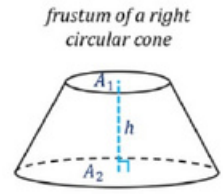
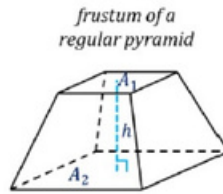
$$LSA = \frac{1}{2} PL$$

- cone

$$V = \frac{1}{3} \pi r^2 h$$

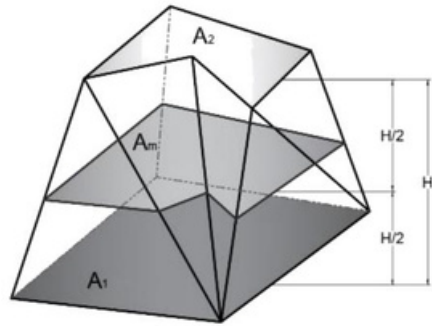
$$TSA = \pi r^2 + \pi r l$$

Frustum



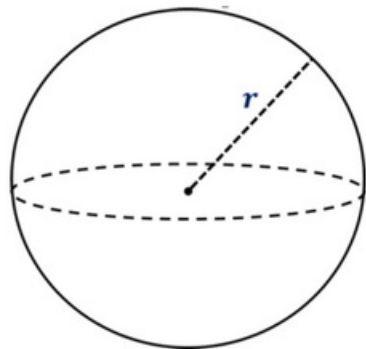
$$V = \frac{h}{3} (A_1 + A_2 + \sqrt{A_1 A_2})$$

Prismatoids



$$V = \frac{H}{6} (A_1 + 4A_m + A_2)$$

Sphere

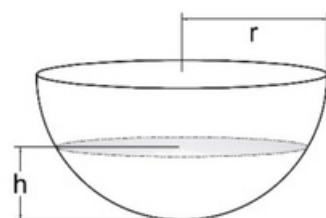


$$V = \frac{4}{3} \pi r^3$$

$$SA = 4\pi r^2$$

Spherical Segments

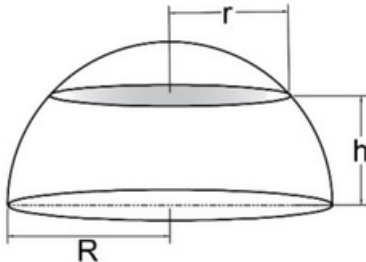
- 1 base



$$V = \frac{\pi h^2}{3} (3r - h)$$

$$A = 2\pi r h$$

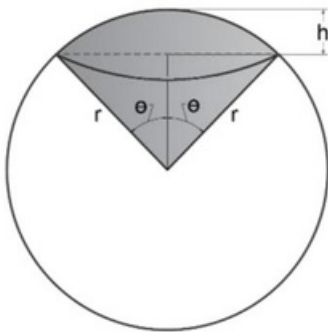
- 2 bases



$$V = \frac{\pi h}{6} (h^2 + 3r^2 + 3R^2)$$

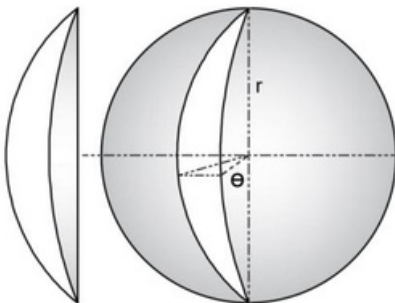
$$A = 2\pi R h$$

Spherical Cone



$$V = \frac{1}{3} A r = \frac{2}{3} \pi r^2 h$$

Spherical Lune & Wedge

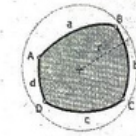


$$A_{lune} = 2r^2 \theta_{rad} = \frac{\pi r^2 \theta}{90}$$

$$V_{wedge} = \frac{2}{3} r^3 \theta_{rad} = \frac{\pi r^3 \theta}{270}$$

Spherical Polygons

A spherical polygon is a polygon on the surface of a sphere whose sides are arcs of great circles.

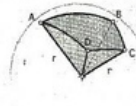


n = number of sides E = spherical excess
 r = radius of sphere

$$\text{Area} = \frac{\pi r^2 E}{180^\circ}$$

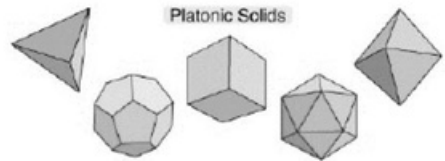
$$E = \text{sum of angles} - (n - 2) 180^\circ$$

Spherical Pyramid



$$\text{Volume} = \frac{\pi r^3 E}{540^\circ}$$

Platonic Solids



PLATONIC SOLIDS

Polyhedra	Vertices	Faces	Edges	Sides	Volume
Cube	8	6	12	Square	e^3
Tetrahedron	4	4	6	Triangle	$0.118e^3$
Octahedron	6	8	12	Triangle	$0.471e^3$
Dodecahedron	20	12	30	Pentagon	$7.663e^3$
Icosahedron	12	20	30	Triangle	$2.182e^3$

- $e = v + f - 2$
- m = number of polygons meeting at a vertex
- radius of circumscribing sphere = $\frac{3V}{As}$

Name	m	SA
Cube	3	$6a^2$
Tetrahedron	3	$a^2\sqrt{3}$
Octahedron	4	$2a^2\sqrt{3}$
Dodecahedron	3	$20.646a^2$
Icosahedron	5	$5a^2\sqrt{3}$

Similar Solids

$$\frac{S_1}{S_2} = \frac{P_1}{P_2} = \left(\frac{A_1}{A_2} \right)^{\frac{1}{2}} = \left(\frac{V_1}{V_2} \right)^{\frac{1}{3}}$$

Unit Conversion

- $1 \text{ m}^3 = 264 \text{ gal}$
- $1 \text{ cm}^3 = 1 \text{ mL}$
- $1 \text{ ton} = 2,000 \text{ lbs}$